Biomolecules

Life comes of chemical reactions in the body running in harmony and in step with one another.

By the end of this unit

- explain what marks out the biomolecules — carbohydrates, proteins, nucleic acids and hormones
- sort the carbohydrates, the proteins, the nucleic acids and the vitamins by structure
- explain how DNA and RNA differ
- describe what the biomolecules do in a biosystem

Where the unit starts from

- a living system **grows, sustains itself and reproduces**, and the most remarkable thing about it is that it is **built of non-living atoms and molecules**
- the chemistry that goes on inside a living system belongs to **biochemistry**
- living systems are made of complex biomolecules — **carbohydrates, proteins, nucleic acids, lipids** and others
- our food must hold **protein and carbohydrate** to be complete
- the biomolecules **act on one another**, and out of that comes the **molecular logic behind life's processes**
- **vitamins and mineral salts**, simple molecules though they are, also matter greatly to how an organism functions

A.1 Carbohydrates

- **plants produce most of the carbohydrates**, a very large group among the organic compounds found in nature; **cane sugar, glucose and starch** are familiar members
- most of them answer to the general formula **$C_x(H_2O)_y$**, and were once taken for **hydrates of carbon**, which is where the name came from
- glucose, $C_6H_{12}O_6$, fits that formula as $C_6(H_2O)_6$
- not everything that fits is a carbohydrate: **acetic acid (CH_3COOH)** fits as $C_2(H_2O)_2$ and is not one, while **rhamnose, $C_6H_{12}O_5$**, is one and does not fit
- their reactions show that they carry **specific functional groups**, so chemically the carbohydrates are defined as **optically active polyhydroxy aldehydes or ketones, or compounds that give such units on hydrolysis**
- those that **taste sweet** are also called **sugars**: the common sugar used at home is **sucrose**

The unit takes six families in turn: **A · Carbohydrates**, **B · Proteins**, **C · Enzymes**, **D · Vitamins**, **E · Nucleic acids** and **F · Hormones**.

- the sugar of **milk** is **lactose**
- carbohydrates also go by the name **saccharides**, from the Greek *sakcharon*, sugar

Classification of carbohydrates

- they are classified by **how they behave on hydrolysis**, and fall into **three broad groups**
- **(i) Monosaccharides** — hydrolysis **takes these no further**: they will not break down into a simpler polyhydroxy aldehyde or ketone
- about **20 monosaccharides occur in nature**; **glucose, fructose and ribose** are common ones
- **(ii) Oligosaccharides** — hydrolysis gives these **between two and ten monosaccharide units**
- they divide again into **disaccharides, trisaccharides, tetrasaccharides** and so on, by **how many units the hydrolysis gives**; **disaccharides are the commonest**
- a disaccharide's two units may be **the same or different**: hydrolysing one molecule of **sucrose** gives **one glucose and one fructose**, while **maltose** gives **two of glucose alone**
- **(iii) Polysaccharides** — carbohydrates that yield **a large number of monosaccharide units on hydrolysis**; **starch, cellulose, glycogen and gums** are examples
- polysaccharides **do not taste sweet**, so they are also called **non-sugars**

- carbohydrates are also sorted into **reducing and non-reducing sugars**
- any carbohydrate that **reduces Fehling's solution and Tollens' reagent** is a **reducing sugar**
- **every monosaccharide, aldose or ketose, is a reducing sugar**

Monosaccharides

- monosaccharides are classified further by the **number of carbon atoms** and by the **functional group** they carry
- one with an **aldehyde group is an aldose**; one with a **keto group a ketose**
- the **count of carbon atoms enters the name too**, as Table 10.A shows

Table 10.A **Different Types of Monosaccharides**

Carbon atoms	General term	Aldehyde	Ketone
3	Triose	Aldotriose	Ketotriose
4	Tetrose	Aldotetrose	Ketotetrose
5	Pentose	Aldopentose	Ketopentose
6	Hexose	Aldohexose	Ketohexose
7	Heptose	Aldoheptose	Ketoheptose

Glucose

- glucose occurs **free in nature and in combined form**
- it is present in **sweet fruits and in honey**, and **ripe grapes** hold a great deal of it
- **Preparation · 1 · From sucrose (cane sugar)** — boil sucrose with **dilute HCl or H_2SO_4 in alcoholic solution** and **glucose and fructose come out in equal amounts**

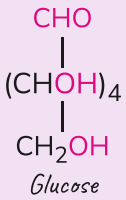
$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

- **2 · From starch** — commercially, glucose comes from **hydrolysing starch**: boil it with **dilute H_2SO_4 at 393 K under pressure**

$$\underset{\text{Starch or cellulose}}{(C_6H_{10}O_5)n} + nH_2O \xrightarrow[393K;\ 2\text{–}3\ atm]{H^+} \underset{\text{Glucose}}{nC_6H_{12}O_6}$$

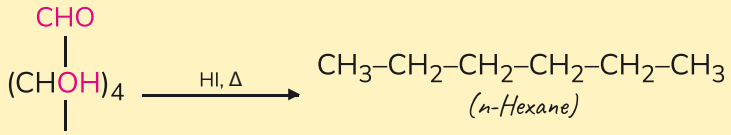
Structure of glucose

- glucose is an **aldohexose**, also known as **dextrose**
- it is the **monomer of many larger carbohydrates**, starch and cellulose among them
- no organic compound on earth is likely to be **more abundant**
- the structure below was assigned on the evidence that follows



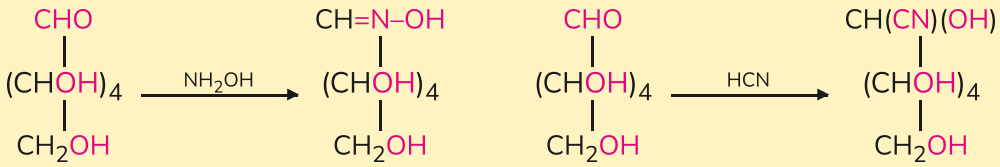
Glucose

- **1** its molecular formula was found to be **$C_6H_{12}O_6$**
- **2** prolonged heating with **HI** gives ***n*-hexane**, which puts all **six carbon atoms in a straight chain**

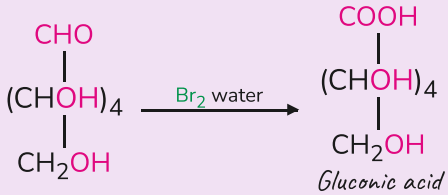


(*n*-Hexane)

- **3** glucose forms an **oxime** with hydroxylamine and adds a molecule of **hydrogen cyanide** to give a **cyanohydrin**, which confirms a **carbonyl group (>C = O)** in it

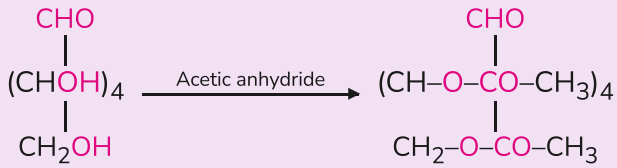


- **4** a **mild oxidising agent** such as **bromine water** oxidises glucose to the **six carbon carboxylic acid, gluconic acid**, which shows the carbonyl group to be an **aldehydic** one

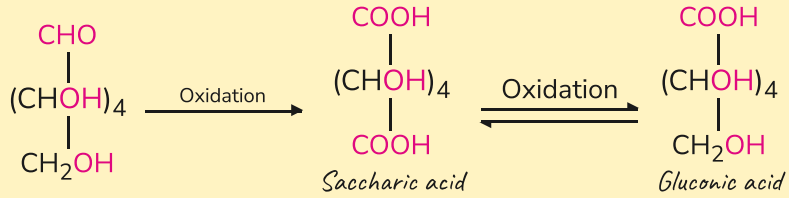


Gluconic acid

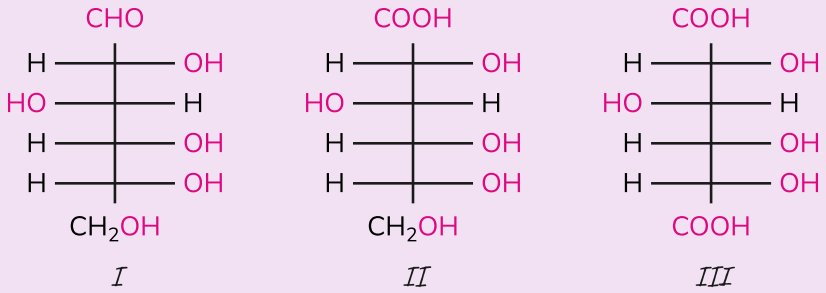
- **5** acetic anhydride **acetylates glucose to the pentaacetate**, which puts **five –OH groups** beyond doubt
- that pentaacetate is a **stable compound**, so those five –OH groups must sit on **different carbon atoms**



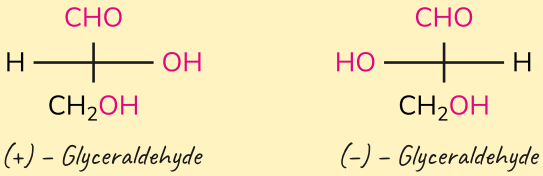
- **6** oxidation with **nitric acid** takes **glucose and gluconic acid alike** to the **dicarboxylic saccharic acid**, which shows a **primary alcoholic (–OH) group** in glucose



- **Fischer** settled the **exact spatial arrangement of the several —OH groups** after studying many further properties
- glucose's configuration is correctly drawn as **I**, so **gluconic acid is II** and **saccharic acid III**

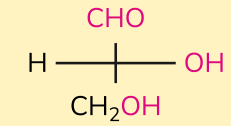


- glucose is correctly named **D(+)-glucose**: the **'D' stands for its configuration** and the **'(+)' for its dextrorotatory nature**
- **'D' and 'L' have nothing to do with optical activity**, and are **unrelated to the letters 'd' and 'l'** (see Unit 6)
- a **'D' or 'L' before a name gives the relative configuration** of that stereoisomer against a compound whose configuration is known
- for carbohydrates that reference is a **particular isomer of glyceraldehyde**, a molecule with **a single asymmetric carbon**, and therefore **two enantiomers**

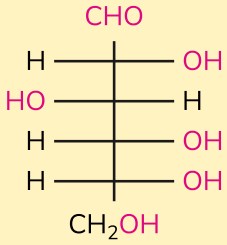


- the **(+) isomer of glyceraldehyde has the 'D' configuration**: written by the settled conventions, its **–OH lies on the right hand side**
- every compound that can be **chemically correlated to D (+) glyceraldehyde** is said to have the **D-configuration**
- every compound correlated to **L (–) glyceraldehyde** has the **L—configuration**, its **–OH on the left hand side**

- for a monosaccharide's configuration it is the **lowest asymmetric carbon atom** that is compared
- in (+) glucose the —**OH on that lowest asymmetric carbon lies on the right**, as it does in (+) glyceraldehyde, so **(+) glucose is assigned the D-configuration**
- the **other asymmetric carbons of glucose take no part** in that comparison
- glucose and glyceraldehyde are both written with the **most oxidised carbon at the top**, here the –CHO



D- (+) - Glyceraldehyde

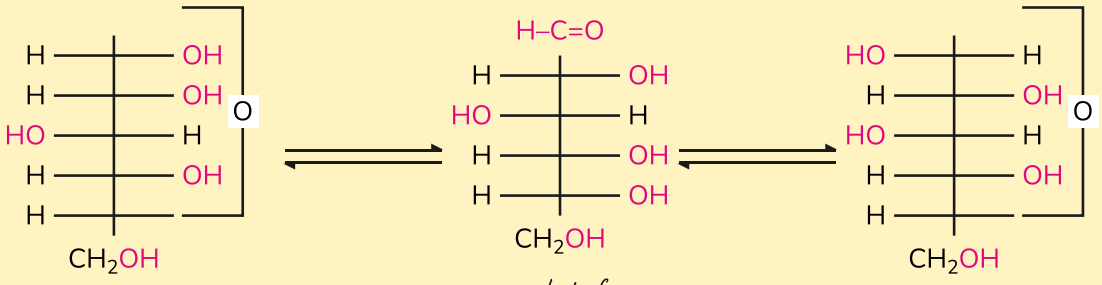


D-(+) - Glucose

Cyclic structure of glucose

- structure **I** explained most of glucose's properties, but **not** the facts below
- **1** for all its aldehyde group, glucose **gives no Schiff's test** and **forms no hydrogensulphite addition product with $NaHSO_3$**
- **2** the **pentaacetate does not react with hydroxylamine**, which says there is **no free —CHO group**
- **3** glucose exists in **two crystalline forms, α and β**; the **α-form (m.p. 419 K)** crystallises from a **concentrated solution at 303 K**, the **β-form (m.p. 423 K)** from a **hot saturated aqueous solution at 371 K**

- the open chain structure (I) could not account for that behaviour
- it was proposed that **one of the —OH groups adds to the —CHO group** and builds a **cyclic hemiacetal**
- glucose was found to form a **six-membered ring**, the **—OH at C-5** taking part in the closure
- that explains both the **absence of a free —CHO group** and the **two forms** below
- the **two cyclic forms stand in equilibrium with the open chain structure**



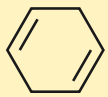
α - D - (+) - Glucose

open chain form

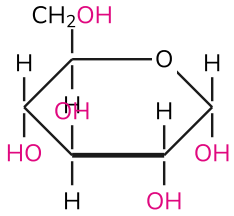
β - D- (+) - Glucose

- between the two cyclic hemiacetal forms **only the hydroxyl group at C1 differs in configuration**; that carbon is the **anomeric** one

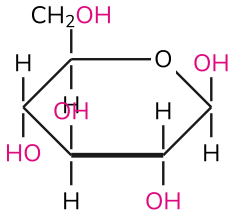
- C1 is the **aldehyde carbon before cyclisation**, and the **α-form and β-form are called anomers**
- glucose's six membered cyclic form is the **pyranose structure**, α– or β–, named after **pyran**, whose ring holds **five carbons and one oxygen**
- the **Haworth structure** represents that cyclic form more correctly



Pyran



α – D – (+) – Glucopyranose



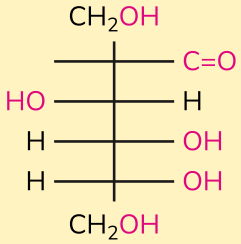
β – D – (+) – Glucopyranose

Fructose

- fructose is an important **ketohexose**; **hydrolysing sucrose** gives it together with glucose
- it is a natural monosaccharide of **fruit, honey and vegetables**, and **pure fructose serves as a sweetner**

Structure of fructose

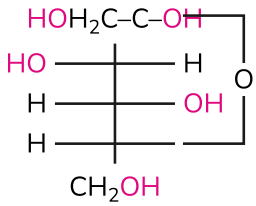
- fructose also answers to $C_6H_{12}O_6$, and its reactions put a **ketonic functional group at carbon number 2** with **six carbons in a straight chain**, as in glucose
- it belongs to the **D-series** and is **laevorotatory**, so it is properly written **D-(–)-fructose**
- it exists in **two cyclic forms** as well, made by **adding the —OH at C5 to the (>C=O) group**
- the ring so formed has **five members** and is the **furanose** form, by analogy with **furan**, a five membered cyclic compound of **one oxygen and four carbon atoms**



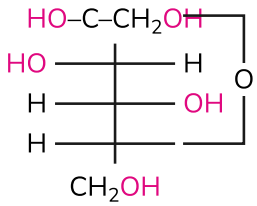
D – (–) – Fructose



Furan

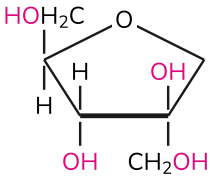


α – D – (–) – Fructofuranose

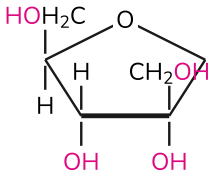


β – D – (–) – Fructofuranose

the two anomers of fructose, drawn as Haworth structures



α – D – (–) – Fructofuranose



β – D – (–) – Fructofuranose

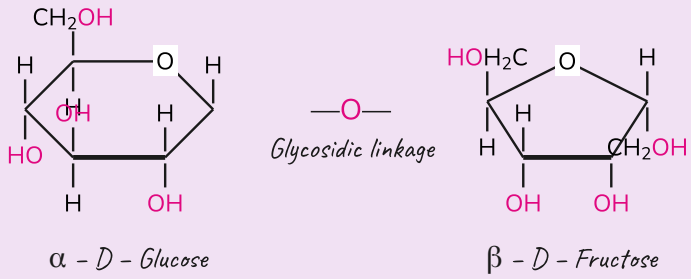
Disaccharides

- a disaccharide hydrolysed by **dilute acid or by enzymes** yields **two monosaccharide molecules**, the same as one another or different
- a **water molecule is lost** and an **oxide linkage** joins the two units; a link of that kind, running through oxygen between two monosaccharides, is a **glycosidic linkage**
- where the **reducing groups of the monosaccharides — the aldehydic or ketonic groups — are bonded**, the disaccharide is **non-reducing**, as sucrose is
- where those groups are **left free**, it is a **reducing sugar**, as maltose and lactose are

- **(i) Sucrose** — one of the common disaccharides; hydrolysis gives an **equimolar mixture of D-(+)-glucose and D-(-) fructose**

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \longrightarrow \underset{\text{D-(+)-Glucose}}{C_6H_{12}O_6} + \underset{\text{D-(−)-Fructose}}{C_6H_{12}O_6}$$

- a **glycosidic linkage from C1 of α-D-glucose to C2 of β-D-fructose** holds the two monosaccharides together
- that linkage ties up the **reducing groups of both**, so **sucrose is a non reducing sugar**

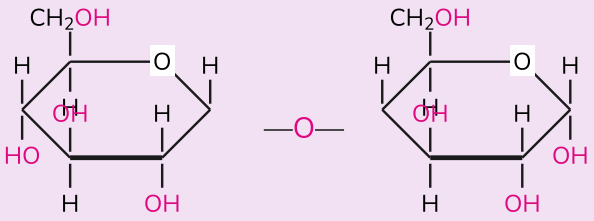


α – D – Glucose β – D – Fructose

Sucrose

- sucrose is **dextrorotatory**, but hydrolysis gives **dextrorotatory glucose and laevorotatory fructose**
- fructose's laevorotation (**–92.4°**) outweighs glucose's dextrorotation (**+ 52.5°**), so the **mixture is laevorotatory**
- so hydrolysis **turns the rotation from dextro (+) to laevo (–)**, and what results is called **invert sugar**

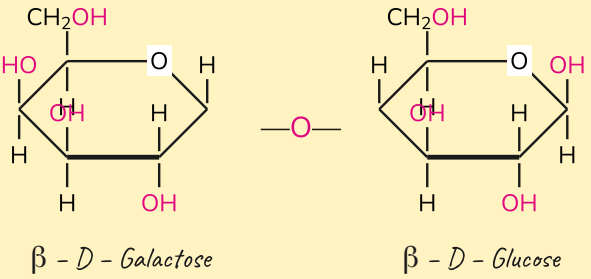
- **(ii) Maltose** — another disaccharide, built of **two α-D-glucose units** in which **C1 of one glucose (I) is linked to C4 of the other (II)**
- in solution a **free aldehyde group can appear at C1 of the second glucose**, so maltose **shows reducing properties** and is a **reducing sugar**



(I) α – D – Glucose (II) α – D – Glucose

Maltose

- **(iii) Lactose** — better known as **milk sugar**, this disaccharide being found in milk
- it is built of **β-D-galactose and β-D-glucose**, the linkage running from **C1 of galactose to C4 of glucose**
- a **free aldehyde group may appear at C-1 of the glucose unit**, so lactose too is a **reducing sugar**

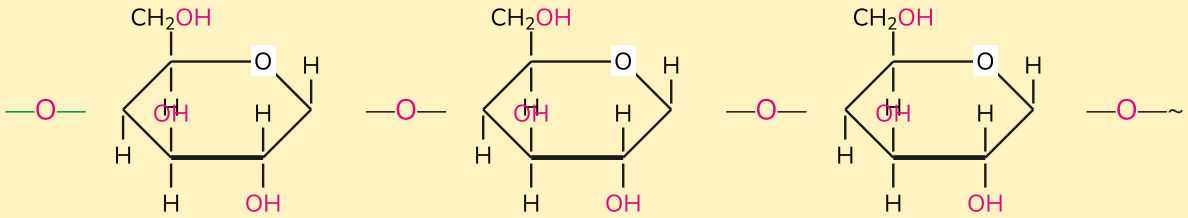


Lactose

Polysaccharides

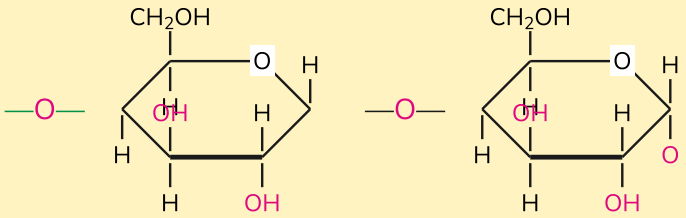
- a polysaccharide holds **a large number of monosaccharide units joined by glycosidic linkages**
- these are the **carbohydrates most commonly met in nature**, and they serve mainly as **food storage or structural materials**

- **(i) Starch** — the **polysaccharide plants store**, and the **most important source of carbohydrate in the human diet**
- it is present in quantity in **cereals, roots, tubers and some vegetables**
- starch is a **polymer of α-glucose** made of two components, **amylose** and **amylopectin**
- **amylose** is the **water soluble** component and makes up about **15-20% of starch**; chemically it is a **long unbranched chain of 200-1000 α-D-(+)-glucose units** held by **C1– C4 glycosidic linkage**
- **amylopectin** is **insoluble in water** and makes up about **80-85%**; it is a **branched chain polymer of α-D-glucose units**, the chain built by **C1–C4 glycosidic linkage** and the **branching by C1–C6**

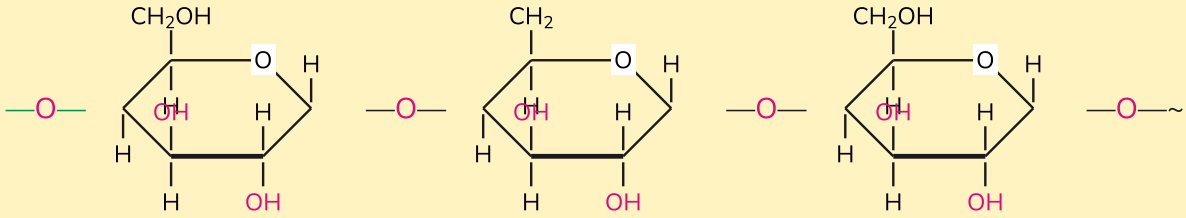


the bridging oxygens are the **α-links** · **Amylose**

Amylopectin

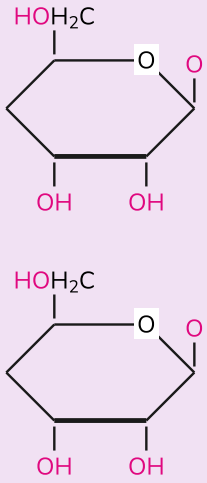


the second unit branches at C_6, which carries the chain below



the bridging oxygens are the **α-links** · **Amylopectin**

- **(ii) Cellulose** — it occurs **in plants alone**, and is the **most abundant organic substance in the plant kingdom**
- the **cell wall of a plant cell is made mostly of it**
- cellulose is a **straight chain polysaccharide, β-D-glucose units and nothing else**, each joined to the next by a **glycosidic linkage running from C1 to C4**



the bridges are the **β-links** · **Cellulose**

- **(iii) Glycogen** — the form in which the **animal body stores carbohydrate**
- it goes by the name **animal starch**, being like amylopectin in structure but **branched a good deal more**
- **liver, muscle and brain** all hold it
- let the body want glucose and **enzymes take the glycogen apart** to supply it
- glycogen is found in **yeast and fungi** as well

Importance of carbohydrates

- plants and animals alike **cannot live without carbohydrates**, and they make up **much of what we eat**
- **honey** has long served as an **instant source of energy**, called '*Vaids*' in the ayurvedic system of medicine
- they are the **storage molecules** — **starch in plants, glycogen in animals**
- **cellulose** makes up the **cell wall of bacteria and of plants**, and from it, in the form of wood, we build furniture — continued on sheet 10

- cotton fibre is cellulose, and it is what we **wear**
- carbohydrates supply raw material to **textiles, paper, lacquers and breweries**
- two aldopentoses, **D-ribose and 2-deoxy-D-ribose** (section E.1), are present in **nucleic acids**
- in a biosystem, carbohydrates are found **combined with many proteins and lipids**

Intext questions

Q1 Glucose and sucrose dissolve in water, while cyclohexane and benzene, simple six membered ring compounds, do not. Explain.

Q2 What products would you expect from hydrolysing lactose?

Q3 How would you account for the absence of an aldehyde group in the pentaacetate of D-glucose?

B.1 Proteins

- no biomolecule of the living system is **more abundant than protein**
- the chief sources are **milk, cheese, pulses, peanuts, fish and meat**
- they are found in **every part of the body**, and **life's structure and its functions rest on them**
- the body needs them to **grow and to keep itself in repair**
- the word protein comes from the Greek **"proteios"**, meaning **primary, or of prime importance**
- **every protein is a polymer of α-amino acids**

Amino acids

- amino acids carry the **amino (–NH_2) and carboxyl (–COOH) functional groups**
- by **where the amino group stands relative to the carboxyl group** they are classified as **α, β, γ, δ** and so on
- **hydrolysing a protein gives α-amino acids only**, and they may carry other functional groups besides
- every α-amino acid has a **trivial name**, usually reflecting a property of the compound or its source
- **glycine** takes its name from its **sweet taste**, Greek *glykos*; **tyrosine** from cheese, Greek *tyros*, where it was first found
- amino acids are generally written as a **three letter symbol**, and sometimes a **one letter** one

R—CH—COOH

α-amino acid (R = side chain)

the —NH_2 group sits on that CH

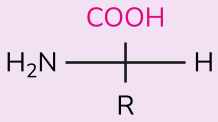


TABLE 10.B **Natural Amino Acids**

Name of the amino acids	Characteristic feature of side chain, R	Three letter symbol	One letter code
1. Glycine	H	Gly	G
2. Alanine	– CH_3	Ala	A
3. Valine*	$(H_3C)_2CH$-	Val	V
4. Leucine*	$(H_3C)_2CH$-CH_2-	Leu	L

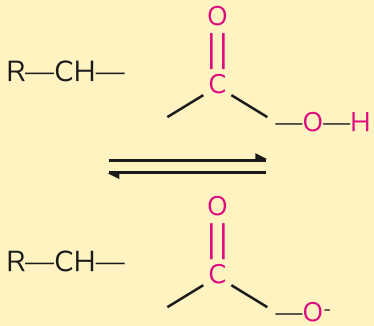
Name of the amino acids	Characteristic feature of side chain, R	Three letter symbol	One letter code
5. Isoleucine*	$H_3C–CH_2–CH(CH_3)–$	Ile	I
6. Arginine*	$HN=C(NH_2)–NH–(CH_2)_3–$	Arg	R
7. Lysine*	$H_2N–(CH_2)_4–$	Lys	K
8. Glutamic acid	$HOOC–CH_2–CH_2–$	Glu	E
9. Aspartic acid	$HOOC–CH_2–$	Asp	D
10. Glutamine	$H_2N–CO–CH_2–CH_2–$	Gln	Q
11. Asparagine	$H_2N–CO–CH_2–$	Asn	N
12. Threonine*	$H_3C–CHOH–$	Thr	T
13. Serine	$HO–CH_2–$	Ser	S
14. Cysteine	$HS–CH_2–$	Cys	C
15. Methionine*	$H_3C–S–CH_2–CH_2–$	Met	M
16. Phenylalanine*	$C_6H_5–CH_2–$	Phe	F
17. Tyrosine	$(p)HO–C_6H_4–CH_2–$	Tyr	Y
18. Tryptophan*	an indole ring, its five membered part carrying –NH, with $–CH_2–$ on the adjacent carbon	Trp	W
19. Histidine*	an imidazole ring carrying $–CH_2–$, with N and NH in the ring	His	H
20. Proline	a five membered ring holding HN and the α-carbon, which carries H and $COOH^a$	Pro	P

* essential amino acid, a = entire structure

Classification of amino acids

- count the **amino groups against the carboxyl groups** in a molecule and it falls out as **acidic, basic or neutral**
- **equal numbers** of the two make it **neutral**
- **more amino than carboxyl groups** makes it **basic**
- **more carboxyl than amino groups** makes it **acidic**
- those the **body can synthesise** are the **non-essential amino acids**
- those it **cannot synthesise**, which must come **through the diet**, are the **essential amino acids**, marked with an asterisk in Table 10.B

- amino acids are usually **colourless crystalline solids**, **soluble in water**, with **high melting points**, and they **behave like salts** rather than like simple amines or carboxylic acids
- the reason: one molecule carries **a carboxyl group, which is acidic, and an amino group, which is basic**
- in water the **carboxyl gives a proton up and the amino group takes one on**, leaving a **dipolar ion, the zwitter ion**
- a zwitter ion is **neutral overall, yet carries both a positive and a negative charge**
- in that form amino acids are **amphoteric**, reacting with acids and with bases alike



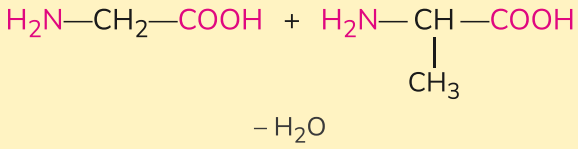
(Zwitter ion)

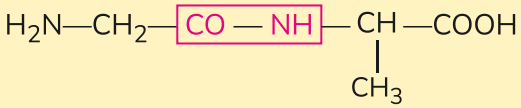
the CH carries :NH_2 on the left, $^{+}NH_3$ on the right

- **glycine excepted**, all naturally occurring α-amino acids are **optically active**, the α-carbon being asymmetric
- they exist in **both 'D' and 'L' forms**, and **most of those found in nature have the L-configuration**
- an L-aminoacid is drawn with the **–NH_2 group on the left hand side**

Structure of proteins

- proteins are **polymers of α-amino acids** connected by a **peptide bond**, or **peptide linkage**
- chemically that linkage is an **amide formed between a –COOH group and an –NH_2 group**
- two amino acid molecules, alike or different, react by **combining the amino group of one with the carboxyl group of the other**
- a **water molecule is eliminated** and a **peptide bond –CO–NH–** forms
- the product is a **dipeptide**, being made of two amino acids: the carboxyl group of **glycine** with the amino group of **alanine** gives the dipeptide **glycylalanine**





Glycylalanine (Gly-Ala) · the boxed part is the peptide linkage

- add a third amino acid to a dipeptide and the **tripeptide** results, **two peptide linkages holding its three amino acids**
- four, five or six linked amino acids give a **tetrapeptide, pentapeptide or hexapeptide**
- **more than ten** makes the product a **polypeptide**
- a polypeptide of **more than a hundred amino acid residues**, with a **molecular mass above 10,000u**, is called a **protein**
- the line between polypeptide and protein is **not sharp**: a polypeptide of fewer amino acids is likely to be called a protein if it has the **well defined conformation** of one, as **insulin** does with its **51 amino acids**

- proteins fall into **two types by molecular shape**
- **(a) Fibrous proteins** — set the **polypeptide chains side by side** and tie them with **hydrogen and disulphide bonds**, and what forms is a **fibre-like structure**
- such proteins are **generally insoluble in water**; **keratin** (in hair, wool and silk) and **myosin** (in muscles) are common examples

(b) Globular proteins

- here the **polypeptide chains coil round on themselves into a sphere**
- water usually dissolves them; **insulin and the albumins** are the common examples
- the structure and shape of proteins are studied at **four levels — primary, secondary, tertiary and quaternary** — each more complex than the one before
- **(i) Primary structure** — a protein may hold **one or more polypeptide chains**; in each, the amino acids are **linked in a specific sequence**, and that **sequence is the protein's primary structure**
- **any change in that sequence creates a different protein**
- **(ii) Secondary structure** — the **shape a long polypeptide chain can take**; two kinds are found, the **α-helix** and the **β-pleated sheet**
- both arise from the **regular folding of the backbone**, brought about by **hydrogen bonding between the >C=O and the –NH– groups of the peptide bond**

Figure 10.A α-Helix structure of proteins

- a single polypeptide chain is drawn **twisted into a right handed screw**, one turn above the next along a straight axis
- the **–NH group of each amino acid residue is hydrogen bonded to the >C=O of an adjacent turn**, the bonds running roughly parallel to the axis
- the ribbon of the drawing stands for the **backbone**, and the hand beside it shows the **right handed sense** of the coil

- the **α-helix** is among the commonest ways for a chain to **make every hydrogen bond it can**: it **twists into a right handed screw**, and the **–NH of each residue bonds to a >C=O one turn away** (Figure 10.A)
- in the **β-pleated sheet** the peptide chains are **drawn out to very nearly their full length**, then **set alongside one another** and tied by **hydrogen bonds between them**
- the result looks like **drapery hanging in pleats**, and that is where the name comes from

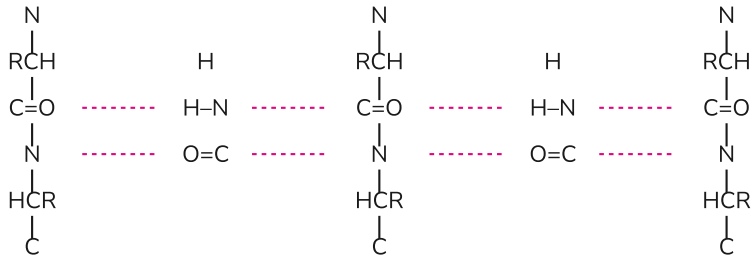


Figure 10.B β-Pleated sheet structure of proteins

- **(iii) Tertiary structure** — **how the polypeptide chains fold overall**, the secondary structure folded once more
- out of it come **two chief molecular shapes: fibrous and globular**
- what steadies the **2° and 3° structures** is **hydrogen bonding, disulphide linkage, van der Waals attraction and electrostatic attraction**

- **(iv) Quaternary structure** — some proteins are built of **two or more polypeptide chains**, called **sub-units**
- the **spatial arrangement of those subunits with respect to one another** is the quaternary structure

- Figure 10.C sets out all four structures diagrammatically, **each coloured ball standing for an amino acid**

Figure 10.C Diagrammatic representation of protein structure (two sub-units of two types in quaternary structure)

- **Primary structure** — a single straight string of beads, one amino acid after another in sequence
- **Secondary structure** — the same string **wound into a regular coil**, turn following turn
- **Tertiary structure** — that coil **folded back on itself** into a compact three-dimensional shape
- **Quaternary structure** — **two such folded sub-units, of two kinds, packed against each other** as one molecule

Figure 10.D Primary, secondary, tertiary and quaternary structures of haemoglobin

- **(a) Primary structure** — a ball and stick fragment of the chain, the atoms marked **C, H, N, O** and the **R groups** hanging off the backbone
- **(b) Secondary structure** — the same chain **coiled as a helix**, the ribbon running through the backbone atoms
- **(c) Tertiary structure** — the helix **folded into one compact globule**, with the **haeme group** set in it
- **(d) Quaternary structure** — **four such globules assembled**, each carrying its own haeme group

Denaturation of proteins

- a protein as it stands in a biological system, with **its own three-dimensional structure and its own biological activity**, is a **native protein**
- subject a native protein to a **physical change**, such as a change of temperature, or a **chemical change**, such as a change of pH, and the **hydrogen bonds are disturbed**
- the **globules unfold and the helix uncoils**, and the protein **loses its biological activity**
- that is the **denaturation** of a protein — continued on sheet 15

- denaturation **wrecks the secondary and the tertiary structure**; the **primary structure survives**
- boil an egg and the **white coagulates**: that is the common example
- another is the **curdling of milk**, brought about by **lactic acid formed by the bacteria present in milk**

Intext questions

Q4 Amino acids melt higher and dissolve in water better than the corresponding halo acids. Explain.

Q5 Where does the water present in an egg go once the egg is boiled?

C.1 Enzymes

- life is possible because the **many chemical reactions in a living organism are coordinated**
- digesting food, absorbing the right molecules and finally producing energy is one such sequence, and **every one of its reactions runs in the body under very mild conditions**
- that happens with the help of **biocatalysts called enzymes**
- **almost all enzymes are globular proteins**
- they are **highly specific**, each to a particular reaction and a particular substrate
- an enzyme usually takes its name from **what it works on**, whether a compound or a class of them: **maltase** catalyses the hydrolysis of maltose to glucose

$$\underset{\text{Maltose}}{C_{12}H_{22}O_{11}} \xrightarrow{\text{Maltase}} \underset{\text{Glucose}}{2\,C_6H_{12}O_6}$$

- sometimes an enzyme is named after the **reaction it is used in**: those catalysing the **oxidation of one substrate with the simultaneous reduction of another** are the **oxidoreductase** enzymes
- the name of an enzyme **ends in -ase**

Mechanism of enzyme action

- enzymes are needed **only in small quantities** to carry a reaction forward
- as chemical catalysts do, enzymes **reduce the magnitude of the activation energy**
- the activation energy for the **acid hydrolysis of sucrose is 6.22 kJ mol^{-1}**, while it is only **2.15 kJ mol^{-1}** when the enzyme **sucrase** does the hydrolysing

D.1 Vitamins

- the **vitamins** are organic compounds our diet must hold **in small amounts**, and want of any one **brings on a specific disease**
- **our body cannot synthesise most of them**, though **plants can make nearly all**, which is why they count as **essential food factors**
- **some of what we need is produced by the bacteria of the gut**
- a normal diet **supplies them all**
- they belong to **many different chemical classes**, so **no structural definition covers them**
- they are taken instead as **organic compounds the diet must supply in small amounts, each for a specific biological function, so that growth and health hold at their best** — continued on sheet 16

- vitamins are designated by the **letters A, B, C, D** and so on; some divide further into **sub-groups such as B_1, B_2, B_6, B_{12}**
- an **excess of vitamins is harmful too**, so **vitamin pills should not be taken without a doctor's advice**
- the term **"Vitamine"** was coined from **vital + amine**, the compounds first identified having **amino groups**
- later work found **no amino group in most of them**, so the **final 'e' went** and the word today is **vitamin**

Classification of vitamins

- vitamins fall into **two groups by their solubility in water or in fat**
- **(i) Fat soluble vitamins — fat and oils dissolve them, water does not**: vitamins **A, D, E and K**
- the body keeps them in the **liver and in adipose, that is fat storing, tissue**

- **(ii) Water soluble vitamins** — the **B group vitamins and vitamin C** dissolve in water and are grouped together
- they **must be supplied regularly in the diet**, being **readily excreted in urine** and **unable to be stored in the body**, vitamin B_{12} excepted

Table 10.C Some important Vitamins, their Sources and their Deficiency Diseases

Sl. No. · Name of Vitamins	Sources	Deficiency diseases
1. Vitamin A	Fish liver oil, carrots, butter and milk	Xerophthalmia (hardening of cornea of eye) · Night blindness
2. Vitamin B_1 (Thiamine)	Yeast, milk, green vegetables and cereals	Beri beri (loss of appetite, retarded growth)
3. Vitamin B_2 (Riboflavin)	Milk, eggwhite, liver, kidney	Cheilosis, the corners of mouth and lips fissuring; digestive disorders; a burning sensation of the skin
4. Vitamin B_6 (Pyridoxine)	Yeast, milk, egg yolk, cereals and grams	Convulsions
5. Vitamin B_{12}	Meat, fish, egg and curd	Pernicious anaemia (RBC deficient in haemoglobin)
6. Vitamin C (Ascorbic acid)	Citrus fruits, amla and green leafy vegetables	Scurvy (bleeding gums)
7. Vitamin D	Exposure to sunlight, fish and egg yolk	Rickets (bone deformities in children) and osteomalacia (soft bones and joint pain in adults)

Sl. No. · Name of Vitamins	Sources	Deficiency diseases
8. Vitamin E	Vegetable oils — wheat germ oil, sunflower oil and the like	Increased fragility of RBCs and muscular weakness
9. Vitamin K	Green leafy vegetables	Increased blood clotting time

E.1 Nucleic acids

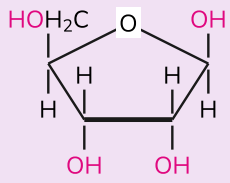
- each generation of a species **takes after its ancestors in many ways**; the question is how those characteristics are passed on
- the **nucleus of a living cell** is responsible for that transmission of inherent characters, which is called **heredity**
- the particles in the nucleus responsible for heredity are the **chromosomes**, made up of **proteins** and of another type of biomolecule, the **nucleic acids**
- nucleic acids are mainly of two kinds: **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**
- they are **long chain polymers of nucleotides**, and so are also called **polynucleotides**

James Dewey Watson

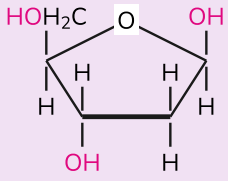
- Dr Watson was born at Chicago, Illinois, in **1928**, and took a **Ph.D. in Zoology at Indiana University in 1950**
- his best known work is the **structure of DNA**; it brought him a share of the **1962 Nobel prize in Physiology and Medicine, with Francis Crick and Maurice Wilkins**
- they put forward the **double helix**: a structure of elegant simplicity, like a **ladder given a gentle twist**
- its **rails alternate phosphate with the sugar deoxyribose**; every **rung is a purine/pyrimidine base pair**
- on that work the new field of **molecular biology** was founded
- because the **nucleotide bases pair complementarily**, the **parental DNA is copied exactly into two daughter cells**
- the revolution in biology that followed led on to **today's recombinant DNA techniques**

Chemical composition of nucleic acids

- hydrolyse DNA or RNA completely and three things come out: a **pentose sugar**, **phosphoric acid**, and **nitrogen containing heterocyclic compounds, the bases**
- in DNA the sugar moiety is **β-D-2-deoxyribose**; in RNA it is **β-D-ribose**

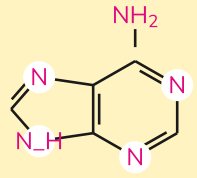


β-D-ribose



β-D-2-deoxyribose

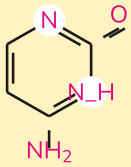
- **DNA holds four bases — adenine (A), guanine (G), cytosine (C) and thymine (T)**
- **RNA holds four as well**: the **first three are the same as in DNA**, but the fourth is **uracil (U)**



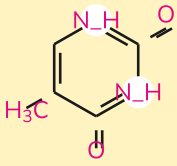
Adenine (A)



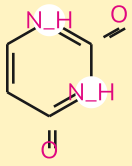
Guanine (G)



Cytosine (C)



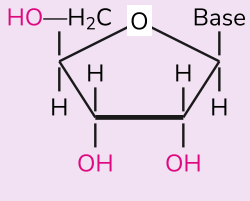
Thymine (T)



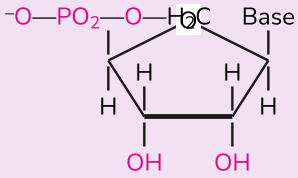
Uracil (U)

Structure of nucleic acids

- a unit formed by **attaching a base to the 1' position of the sugar** is a **nucleoside**
- in a nucleoside the **sugar carbons are numbered 1', 2', 3'** and so on, to distinguish them from the bases (Figure 10.Ea)
- **link that nucleoside to phosphoric acid at the 5'-position of the sugar moiety** and a **nucleotide** results (Figure 10.Eb)



(a)



(b)

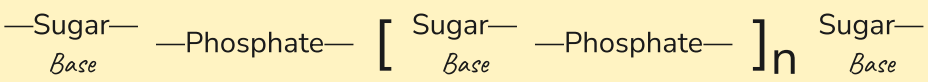
Figure 10.E Structure of (a) a nucleoside and (b) a nucleotide

- nucleotides are joined by a **phosphodiester linkage between the 5' and 3' carbon atoms of the pentose sugar**
- sheet 19 shows how a typical **dinucleotide** is formed

Figure 10.F Formation of a dinucleotide

- **Left** — two separate nucleotides, each drawn as a **phosphate block** joined through –O– to a **CH_2**, then to a **sugar ring** that carries its **base** at 1' and a free **–OH at 3'**
- **Right** — the **3'-OH of the upper sugar and the phosphate of the lower nucleotide have condensed**, leaving a **phosphodiester linkage** between them
- the chain so built runs from a **5' end**, where the free phosphate is, to a **3' end**, where the free –OH is

- a nucleic acid chain is written in simplified form as below



- a nucleic acid's **primary structure** is the **order its nucleotides stand in along the chain**
- they have a **secondary structure** as well: the **double strand helix** that **James Watson and Francis Crick** gave DNA (Figure 10.G)
- **two nucleic acid chains wind about one another**, held by **hydrogen bonds from base to base**
- the strands are **complementary**, since only **certain bases pair**: **adenine to thymine** and **cytosine to guanine**
- in the secondary structure of **RNA a single stranded helix** is present, which **sometimes folds back on itself**
- there are **three types of RNA**, each with its own work: **messenger (m-RNA), ribosomal (r-RNA) and transfer (t-RNA)**

Figure 10.G Double strand helix structure for DNA

- two ribbons **wind round a common axis**, one running **5' to 3'** and the other **3' to 5'**, so the strands lie **antiparallel**
- the rungs between them are the **base pairs**, written **A=T** where two hydrogen bonds hold, and **C≡G** where three do

Har Gobind Khorana

- born in **1922**, he read for an **M.Sc. at Punjab University, Lahore**
- under **Professor Vladimir Prelog** his thinking, and his whole view of **science, of work and of effort**, were shaped
- a short spell in India in **1949** over, he went back to England, to **Professor G.W. Kenner and Professor A.R.Todd**
- **Cambridge** is where his interest in **proteins and nucleic acids** began
- for **cracking the genetic code** he took a share of the **1968 Nobel Prize for Medicine and Physiology, with Marshall Nirenberg and Robert Holley**

DNA Fingerprinting

- every individual has **unique fingerprints**, found at the tips of the fingers and long used for identification, though **surgery can alter them**
- a **sequence of bases on DNA is also unique to a person**, and the information about it is called **DNA fingerprinting**
- it runs the **same in every cell**, and **no treatment known will change it**
- it is now used **(i)** in forensic laboratories to **identify criminals**; **(ii)** to **determine paternity**; **(iii)** to **identify dead bodies in an accident**, by comparing the DNA of parents or children; **(iv)** to **identify racial groups and rewrite biological evolution**

Biological functions of nucleic acids

- heredity rests chemically on DNA, which may be thought of as **where the genetic information is held in reserve**
- it alone has **kept the several species of organisms true to themselves across millions of years**
- a DNA molecule can **duplicate itself as the cell divides**, so the **daughter cells receive identical strands**
- nucleic acids have another important office: the **synthesis of protein in the cell**
- the **RNA molecules do the synthesising**, but the **instruction for any particular protein lies in the DNA**

F.1 Hormones

- a hormone is a molecule that carries **messages between cells**
- they are **produced by endocrine glands** and **poured directly into the blood stream**, which carries them to the **site of action**
- chemically they vary: **estrogens and androgens are steroids**; **insulin and the endorphins are poly peptides**; **epinephrine and norepinephrine derive from amino acids**

- hormones **hold the body's biological activities in balance**
- **insulin** keeping the blood glucose level within a narrow limit is one example of that work: it is **released in response to a rapid rise in blood glucose**
- **glucagon**, on the other hand, tends to **raise the glucose level in the blood**, and the two hormones together **regulate it**
- **epinephrine and norepinephrine** carry the **body's response to a stimulus from outside**
- **growth hormones and sex hormones** have their part in **how the body grows and develops**
- the **thyroid gland** produces **thyroxine**, which is the amino acid **tyrosine iodinated**; too little of it brings the trouble set out on sheet 21

- too little thyroxine brings **hypothyroidism**, marked by **lethargyness and obesity**; too much brings **hyperthyroidism**
- let the **diet run short of iodine** and hypothyroidism may follow, with the **thyroid gland enlarging**
- the condition is largely held in check by **putting sodium iodide into commercial table salt**, which is then sold as "Iodised" salt

- **steroid hormones** are produced by the **adrenal cortex** and the **gonads** — testes in males, ovaries in females
- those from the adrenal cortex play a very important part in the body's functions: **glucocorticoids control carbohydrate metabolism, modulate inflammatory reactions and take part in reactions to stress**
- **mineralocorticoids control how much water and salt the kidney excretes**

- an adrenal cortex that fails to work properly may leave **Addison's disease**, which brings **hypoglycemia, weakness, and a body less able to bear stress**
- it **kills unless glucocorticoids and mineralocorticoids are given**
- the **secondary sex characters develop under the hormones the gonads release**
- the male's chief sex hormone is **testosterone**, which brings the **deep voice, the facial hair and the general physical constitution**
- **estradiol** is the main female sex hormone, responsible for the **secondary female characteristics** and taking part in the **control of the menstrual cycle**
- **progesterone** prepares the **uterus for implantation of the fertilised egg**

Intext questions

Q6 Why can our body not store vitamin C?

Q7 Hydrolyse a nucleotide from DNA that contains thymine. What products would form?

Q8 Hydrolysing RNA leaves no relationship among the quantities of the different bases obtained. What does that suggest about the structure of RNA?

Summary

- a **carbohydrate** is an **optically active polyhydroxy aldehyde or ketone**, or a molecule that **hydrolyses to such units**; broadly they are **monosaccharide, disaccharide or polysaccharide**
- **glucose** matters most to a mammal for energy, and **digesting starch** yields it; **glycosidic linkages** tie monosaccharides into disaccharides and polysaccharides
- a **protein** is a polymer of **some twenty different α-amino acids**, tied by **peptide bonds**
- **ten of them count as essential**: our body cannot make them, and the **diet must supply them**
- in an organism proteins do work both **structural and dynamic**; one built of **α-amino acids alone** is a **simple protein**
- the **secondary or tertiary structure of a protein is disturbed by a change of pH or of temperature**, and the protein can no longer perform its functions: that is the **denaturation of proteins**
- **enzymes** are the **biocatalysts** that hurry a biosystem's reactions along; their action is **highly specific and selective**, and **chemically they are for the most part proteins**
- **vitamins** are the **accessory food factors a diet must carry**, sorted into **fat soluble — A, D, E, K** — and **water soluble — the B group with C**; go short of them and **many diseases follow**

- **nucleic acids are polymers of nucleotides**, and a nucleotide in turn is made of a **base, a pentose sugar and a phosphate moiety**
- they are what **carries characters from parent to offspring**
- there are **two types — DNA and RNA**: DNA holds the five carbon sugar **2-deoxyribose**, RNA holds **ribose**
- **adenine, guanine and cytosine belong to both**; for the fourth base DNA has **thymine** and RNA **uracil**
- DNA runs as a **double strand**, RNA as a **single** one
- heredity rests chemically on DNA, which holds **in code the message for every protein the cell must synthesise**
- **mRNA, rRNA and tRNA** are the three types, and **they do the synthesising**

Exercises

E1 What are monosaccharides?

E2 What are reducing sugars?

E3 Give two main functions of carbohydrates in plants.

E4 Which of ribose, 2-deoxyribose, maltose, galactose, fructose and lactose are monosaccharides, and which disaccharides?

E5 What does the term glycosidic linkage mean to you?

E6 What is glycogen, and how does it differ from starch?

E7 What does hydrolysis give from (i) sucrose and (ii) lactose?

E8 In structure, what basically separates starch from cellulose?

E9 What happens when D-glucose is treated with each of these reagents? (i) HI; (ii) Bromine water; (iii) HNO_3

E10 List the reactions of D-glucose that its open chain structure cannot explain.

E11 Explain the terms essential and non-essential amino acid, and name two of each.

E12 Give a definition, in the context of proteins, of the peptide linkage, of primary structure, and of denaturation.

E13 Which secondary structures do proteins commonly take?

E14 What type of bonding helps stabilise the α-helix structure of proteins?

E15 Set out the difference between globular and fibrous proteins.

E16 How do you account for the amphoteric behaviour of amino acids?

E17 What are enzymes?

E18 What does denaturation do to the structure of proteins?

E19 On what basis are vitamins classified, and which one is behind the coagulation of blood?

E20 Why must we have vitamin A and vitamin C, and where do they chiefly come from?

E21 What are nucleic acids? Mention two of their important functions.

E22 How does a nucleoside differ from a nucleotide?

E23 DNA's two strands are complementary rather than identical. Explain.

E24 Where do DNA and RNA differ, in structure and in function?

E25 Which types of RNA occur in the cell?